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Numerical computing HW#1

1.1: 2, 10, 11, 14

2) Show Solutions exist on Interval

a) $\sqrt{x} - \cos x = 0$ on $[0,1]$

$f(0) = \sqrt{0} - \cos(0) = -1$

$f(1) = \sqrt{1} - \cos(1) \approx 1 - 0.54 = .46$

-Since function is continuous and the values of the function go from NEGATIVE to POSITIVE, there must be a root

b) $e^x - x^2 + 3x - 2 = 0$ on $[0,1]$

$f(0) = e^0 - \cancel{0^2} + \cancel{3(0)} - 2 = 1 - 2 = -1$

$f(1) = e^1 - (1)^2 + 3(1) - 2 \approx e$

Funct must have a root for the same reason, continuous and switches signs

c) $-3\tan(2x) + x = 0$ on $[0,1]$

$f(0) = -3\tan(0) + 0 = 0$

$f(1) = -3\tan(2) + 1 = 7.56$

Root is Given by $f(0)$ on interval. Asymptote is @ $\frac{\pi}{4}$ so IVT wouldn't Apply

d) $\ln x - x^2 + \frac{5}{2}x - 1 = 0$ on $[\frac{1}{2}, 1]$

$f(\frac{1}{2}) = \ln(\frac{1}{2}) + (\frac{1}{2})^2 + \frac{5}{2}(\frac{1}{2}) - 1 = -0.69$

$f(1) = \ln(1) + (1)^2 + \frac{5}{2}(1) - 1 = \frac{1}{2}$

Root exist, function is continuous and signs change on either end of interval

10) Find Taylor Polynomial $P_3(x)$ for $f(x) = \sqrt{x+1}$, $x_0 = 0$

Approx $\sqrt{.5}$, $\sqrt{.75}$, $\sqrt{1.25}$, $\sqrt{1.5}$

$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!}$$

$$= 1 + \frac{1}{2}x + \frac{1}{2!\cdot 4}x^2 + \frac{3}{3!\cdot 8}x^3$$

$$\boxed{P_3(x) = 1 + \frac{1}{2}x + \frac{x^2}{8} + \frac{x^3}{16}}$$

$f(0) = 1$

$f'(x) = \frac{d}{dx}[(x+1)^{1/2}] = \frac{1}{2}(x+1)^{-1/2}$

$= \frac{1}{2\sqrt{x+1}}$ $\quad f'(0) = \frac{1}{2}$

$f''(x) = \frac{d}{dx}[\quad] = -\frac{1}{4}(x+1)^{-3/2}$

$= -\frac{1}{4(x+1)^{3/2}}$ $\quad f''(0) = -\frac{1}{4}$

$f'''(x) = \quad \frac{3}{} \quad f'''(0) = \frac{3}{}$

10) cont. Approx $\sqrt{.5}, \sqrt{.75}, \sqrt{1.25}\ \sqrt{1.5}$

$P_3(x) = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}$ is for $\sqrt{x+1}$ abt $x_0 = 0$

$P_3(-.5) = 1 + \frac{(-.5)}{2} - \frac{(-.5)^2}{8} + \frac{(-.5)^3}{16}$

$= 1 + 0.25 + 0.125 - 0.0078125 = 0.7109375 \approx \sqrt{.5}$

$P_3(-.25) = 1 + \frac{(-.25)}{2} - \frac{(-.25)^2}{8} + \frac{(-.25)^3}{16} = 0.866210930 \approx \sqrt{.75}$

$P_3(.25) = 1 + \frac{(.25)}{2} - \frac{(.25)^2}{8} + \frac{(.25)^3}{16} = 1.1181640625 \approx \sqrt{1.25}$

$P_3(.5) = 1 + \frac{.5}{2} - \frac{(.5)^2}{8} + \frac{(.5)^3}{16} = 1.2265625 \approx \sqrt{1.5}$

11) $P_2(x)$ for $f(x) = e^x \cos x \quad x_0 = 0$

$f(0) = e^0 \cos 0 = 1$

$f'(x) = e^x \cos x - e^x \sin x \Rightarrow f(0) = 1 - 0 = 1$

$f''(x) = -2e^x \sin x \Rightarrow f(0) = 0$

$P_2(x) = f(0) + f'(0)x + \frac{f''(c)x^2}{2!} \Rightarrow \boxed{P_2(x) = 1 + x}$

a) $P_2(.5) = 1 + .5 = \underline{1.5 \approx f(.5)}$ Approx

$|f(.5) - P_2(.5)|$

$\hookrightarrow E_3 = \frac{f'''(\xi)x^3}{3!}$ using $\xi = .5$

$E_3 = \frac{-2e^{\xi}(\sin\xi + \cos\xi)x^3}{3!}$ so $\boxed{|f(.5) - P_3(.5)| \leq .0932}$

b) Find Bound for $|f(x) - P_2(x)|$ using $P_2(x)$ to Approx $f(x)$ on $[0,1]$

$f(0) = e^0 \cos(0) = 1 \qquad P_2(0) = 1$

$f(1) = e^1 \cos(1) = e + 1 \qquad P_2(1) = 2$

$|f(x) - P_2(x)| = |e + 1 - 2|$ so $\boxed{|f(x) - P_2(x)| \leq 1.718}$

c) Approx $\int_0^1 f(x)dx$ using $\int_0^1 P_2(x)dx$

$\int_0^1 1 + x\, dx = 1.5 \leftarrow$ From A so $\int_0^1 f(x) \approx 1.5$

d)

11)

d) Find Upper Bound for error in (c) using $\int_0^1 |R_2(x)\,dx|$
compare to bound of actual Error

$\left|\int_0^1 f(x)-P_2(x)\right| \le \int_0^1 |R_2(x)\,dx| \le\ ? \Rightarrow \left|\int_0^1 f(x)-P_2(x)\right| \le \int_0^1 |R_2(x)\,dx| \le .3$

$\int_0^1 R_2(x) = \int_0^1 \frac{-2e^x(\sin x+\cos x)\,x^3}{3!} = -2e^x \sin x\Big]_0^1 =$
$= -2e^1 \sin(1) - 0$
≈ 0.313

Actual error is 0.122

14) $f(x) = 2x\cos(2x) - (x-2)^2 \quad x_0 = 0$

a) find $P_3(x)$ approx $f(.4)$

$f'(x) = -2x + 2(-2x\sin(x) + \cos(2x)) + 4 \qquad f(0) = -4$
$f''(x) = 2(-4x\cos(2x) - 4\sin(2x)) - 2 \qquad f'(0) = 6$
$f^3(x) = 2(8x\sin(2x) - 12\cos(2x)) \qquad f''(0) = -2$
$f^3(0) = -24$

$P_3(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f^3(0)x^3}{3!} + \underbrace{\frac{f^{(4)}(\xi(x))x^4}{4!}}_{\text{ERROR}}$

$= -4 + 6x - \frac{2x^2}{2} - \frac{24x^3}{3!}$

$\boxed{P_3(x) = -4 + 6x - x^2 - 4x^3}$

$f(.4) \approx P_3(.4) = -4 + 6(.4) - (.4)^2 - 4(.4)^3$
≈ -2.016

b) Error Formula to find Upper Bound for $|f(.4) - P_3(.4)|$

$|R_3(.4)| = \frac{f^4(0)x^4}{4!}$, $\quad f^4(x) = 2(16x\cos(2x) + 32\sin(2x))$
$f^4(0) = 0$

$|R_3(.4)| = \frac{(.4)^4}{24} \le .00106$

$f(.4) = -2.00263$ SO $|f(.4) - P_3(.4)| \le 0.0726$

c) $P_4(x) = P_3(x)$ bc $f^4(0) = 0$ so no term is added
error becomes $\frac{f^{(5)}(\xi(x))x^5}{5!}$

d) Error is the Same
$|f(.4) - P_4(.4)| \le 0.0726$